TEAM ASTRO

AeroSentinel

Satellite recovery and mission planning

An evaluation environment for tool-using agents

2une.ai RL Hackathon



The problem: a repair can arrive too late

A satellite has limited power,
short contact windows,
and data it cannot afford to lose.

Time

Energy

Storage

Reading a manual is only the start. The agent must decide what to change and when to act.

AeroSentinel scores the mission history.

How an episode works

01

Observe

Flight manual
Configuration files
Live telemetry

02

Act

Inspect and edit files
Run Python
Advance the orbit

03

Evaluate

Persistent physical state
Protected mission targets
Reward and final checks

40 replies

10 tools per reply

600-second episode

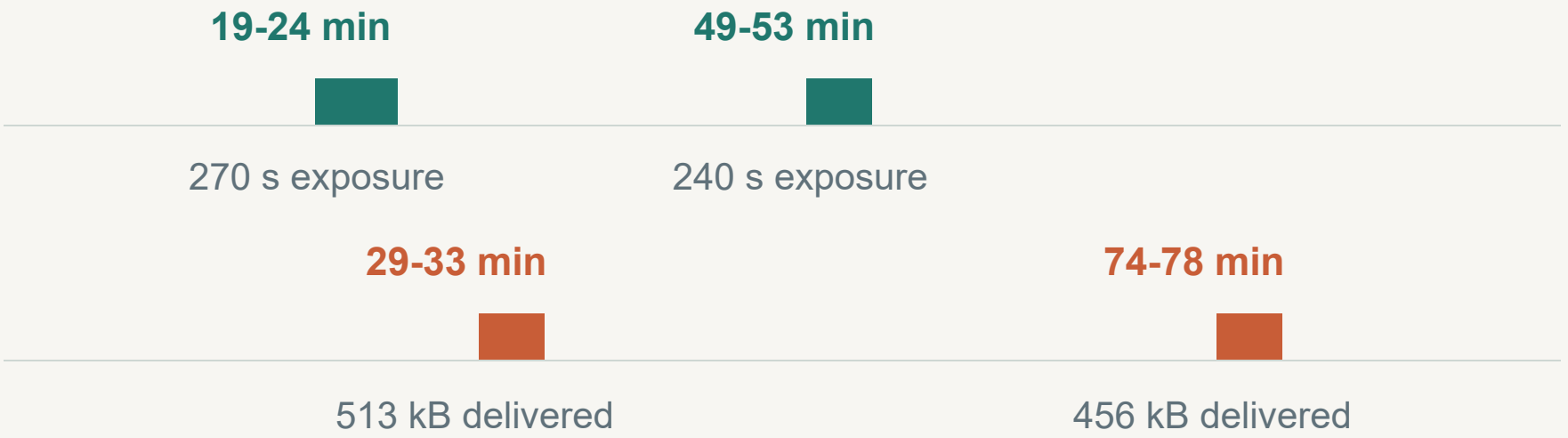
Nine tasks, with increasing planning depth

Task group	Families	What the agent must decide
Guided recovery	Power, pointing, thermal, radio and fault cascade	Diagnose a fault and repair it
Radio planning	RF energy tradeoff Early pass recovery	Balance energy and link rate Repair before contact closes
Payload planning	Observation scheduling Science store and forward	Collect at the right time Manage storage and downlink

3 difficulty levels. Seeded faults, hardware and timing.

The planning test: collect, transmit, repeat

Science mission, seed 101



The first contact frees the buffer for the second acquisition.

969 kB delivered **0 kB lost** 598.5 kB buffer

What earns a successful score

Reward measures progress from the seeded no-op baseline.

Progress = clamp((raw - no-op) / (1 - no-op), 0, 1)

Reward ≥0.95 requires every mandatory check.

A missed science window, buffer overflow or invalid control keeps the run below the success threshold.

The agent cannot award itself success

Protected verifier
Separate container
Bounded tool execution

Six deliberate failure probes all stay below success.

Executed probe	Reward
Relaxed pointing target	0.800
Forged telemetry	0.000
Missed early contact	0.418
Missed observation	0.800
Missed science contact	0.609
Forged science inventory	0.521

Docker and the host kernel remain trusted.

Validation on the final source

474

mission expectations passed

80 feasible plans succeed.
394 invalid choices are rejected.

Check	Observed result
Linux regression tests	150 passed
Actual-container isolation	12 passed
Reference / no-op gates	9 / 9 passed
Clean Docker build	13.77 s of 180 s allowed

GitHub CI passed on the submitted main commit.

What the organizer model actually achieved

gpt-6-luna, using the same model interface

Recorded attempt	Result	Reward	Replies
EPS recovery, no append	Success	1	11
RF energy, seed 101	Success	1	11
Science, seed 101	Success	1	28
Observation, seed 101	Failure	0.8	12

Three successes. One planning failure, preserved in the evidence.

Four attempts on one deployment. Unknown judge append instructions remain untested.

Demo: EPS eclipse recovery

Inspect the fault. Apply a repair.
Show the final mission checks.

Recorded Run A

Reward 1.0

11 replies

Current limits

Simplified physics. Finite exploit coverage.
One organizer-model deployment tested.